

Lab 17: Ecology — Bioaccumulation & Biomagnification of DDT

BIOL-8

Name: _____ Date: _____

Purpose

Bioaccumulation and biomagnification of DDT: You will observe how a contaminant accumulates in organisms and becomes more concentrated at higher levels of a simple food web, using beads to model producers, fish, and a fish-eating bird.

Learning objectives

- Explain **bioaccumulation** (buildup of a contaminant within an individual) and **biomagnification** (increasing concentration at higher trophic levels).
 - Track **DDT bead** counts through a model food chain: producers → stickleback → cutthroat trout → osprey.
 - Interpret bead data to describe how **DDT** moves and concentrates in a simplified aquatic web.
 - Connect **DDT** to documented effects on **top predators** (e.g., eggshell thinning in birds) and to **indirect** effects on other populations when a predator declines.
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Materials

- **Shaker cup (plastic)** — holds the producer population
- **9 small cups** — stickleback fish
- **3 medium cups** — cutthroat trout
- **1 large cup** — osprey
- **20 beads of one color** — producer biomass
- **10 beads of a second color** — DDT (pesticide)

Total: 30 beads in the shaker at the start, with **DDT making up 1/3** of the starting mixture (10 of 30 beads).

Background

This lab explores **bioaccumulation** (buildup of contaminants within an individual organism) and **biomagnification** (increasing concentration at higher trophic levels), using **DDT** as the case study. DDT was inexpensive to manufacture and persistent in the environment. It enters aquatic systems by binding to plankton, accumulates in their cells, passes to fish that eat the plankton, and then to birds that eat the fish. In birds, DDT disrupts **eggshell formation**, producing brittle shells that break before hatching.

Procedure

Setup

1. Place all **30 beads** (20 producer + 10 DDT) into the **shaker cup**.
2. In the data table, record the **DDT-to-total ratio** in the shaker ($10/30 = 1/3$). Shake thoroughly to mix.

Trophic level 1 → 2 (producers → stickleback)

1. Remove **3 beads** from the shaker and place them into **each** of the **9** stickleback cups (27 beads moved).
2. Record the **number of DDT beads** in **each** stickleback cup.

Trophic level 2 → 3 (sickleback → cutthroat trout)

1. Each trout eats **2 sickleback** — empty **2 small cups** into **each** of the **3** medium (trout) cups.
2. Record **DDT per trout**.

Trophic level 3 → 4 (trout → osprey)

1. Each osprey eats **2 trout** — empty **2 medium cups** into the **large** (osprey) cup.
2. Record **DDT in the osprey**.

Trials and calculations

1. Repeat the **full procedure** for a total of **3 trials**.

2. Calculate the **average** and **total** DDT for each organism where your instructor directs (e.g., per trophic level or per cup type).
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Data

Trial 1

Initial mixture (shaker): DDT bead count / 30 total beads; fraction DDT = (expected about **1/3** by count, or 10/30)

Trial 1 — Stickleback cups (DDT beads per cup)

#	Cup	DDT beads
1		
2		
3		
4		
5		
6		
7		
8		
9		

Trial 1 — Trout and osprey

#	Organism	Cup / role	DDT beads
1			
2			
3			
4			

Trial 2

Initial mixture (shaker): DDT bead count / 30; fraction
DDT =

Trial 2 — Stickleback cups (DDT beads per cup)

#	Cup	DDT beads
1		
2		
3		
4		
5		
6		
7		
8		
9		

Trial 2 — Trout and osprey

#	Organism	Cup / role	DDT beads
1			
2			
3			
4			

Trial 3

Initial mixture (shaker): DDT bead count / 30; fraction
DDT =

Trial 3 — Stickleback cups (DDT beads per cup)

#	Cup	DDT beads
1		
2		
3		
4		
5		
6		
7		
8		
9		

Trial 3 — Trout and osprey

#	Organism	Cup / role	DDT beads
1			
2			
3			
4			

Summary — averages across trials

Summary — Average or total DDT (as directed)

#	Metric	Trial 1	Trial 2	Trial 3	Average across trials
1					
2					
3					
4					

Analysis questions

Answer in **complete sentences**. Refer to your **data tables** where a question asks for numbers or patterns.

1. For each of the three trials, calculate the **average DDT per organism** (as defined by your instructor for stickleback, trout, and osprey).

2. Construct a **food chain** illustrating the flow of energy through this model (from producers through osprey). Label trophic levels if asked.

3. What happened to the **amount of DDT** as you moved **up** the food chain?

4. Why is **DDT harmful** to the osprey?

5. Name **another organism** (besides osprey) you would expect to have **high DDT concentrations**, and explain why.

6. If the **osprey population declines** due to DDT, name **two other populations** that would be affected and describe **how**.

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Lab 17 — Ecology. Bioaccumulation and biomagnification using a bead model.